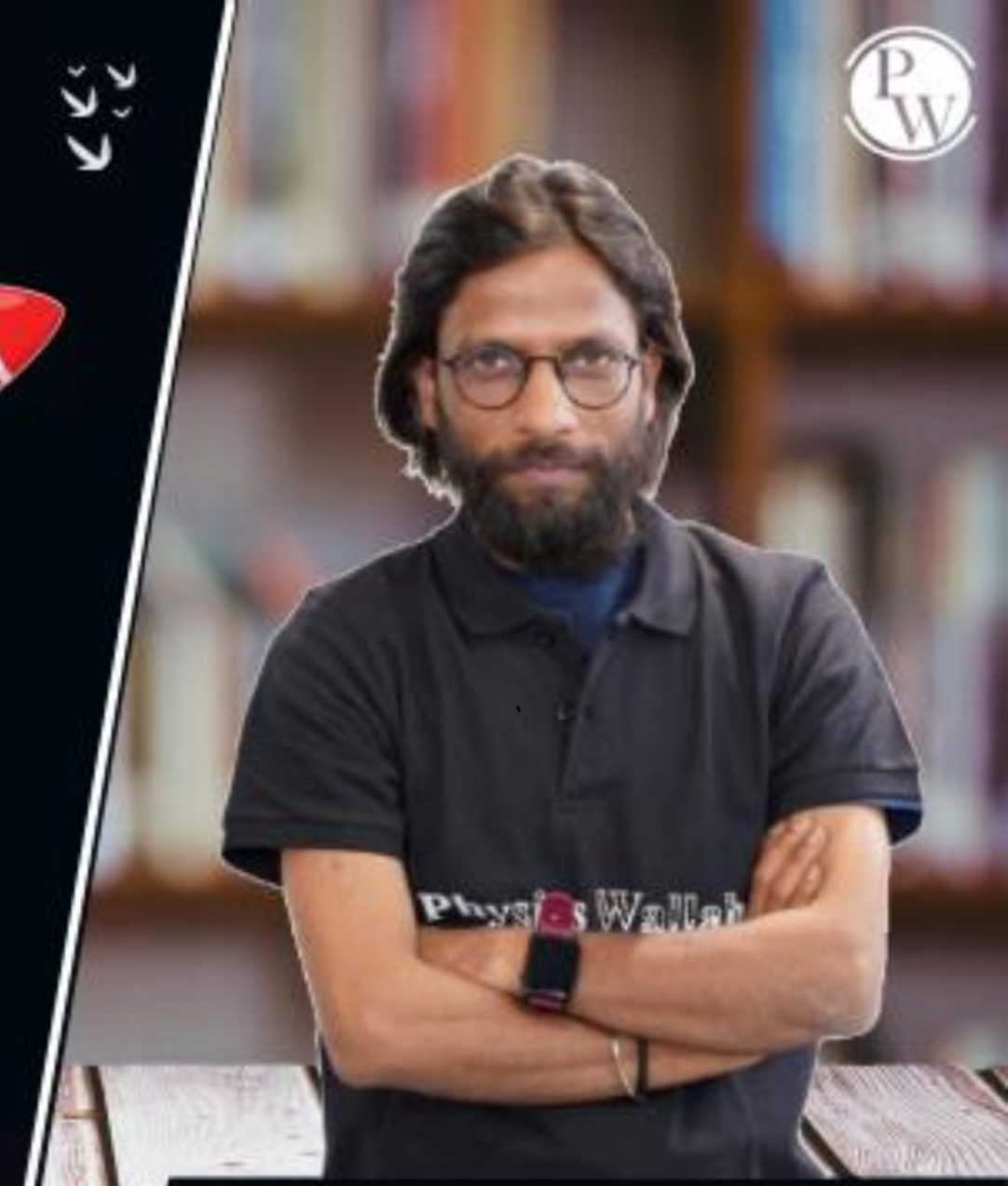
GATE-ALL BRANCH

Engineering Mathematics



PROBABILITY AND STATISTICS

Lecture No.- 04



By- Rahul sir

Recap of Previous Lecture



Topic

Concepts and problems based on fundamental probability

H.W ⟶ $\frac{65}{156}$

Topics to be Covered



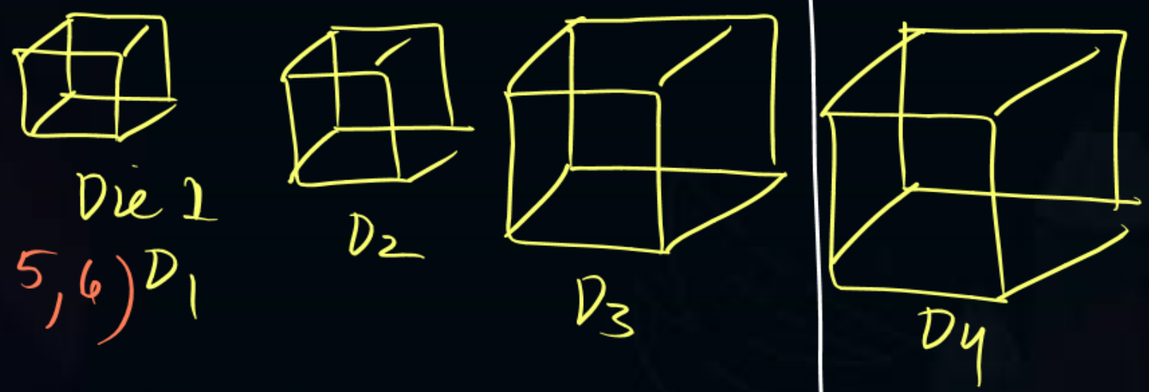
Topic	Classification of events.

Topic : Fundamentals of Probability

Rahul Sir PW

#Q. Four fair dice D_1, D_2, D_3, D_4, each having six faces numbered 1, 2, 3, 4, 5, 6 are rolled simultaneously. The probability that D_4 shows a number appearing on one of D_1, D_2 and D_3 is

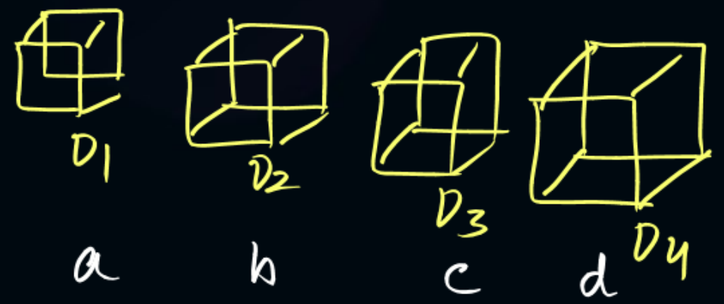
A. 91/216
B. 108/216
C. 125/216
D. 127/216



$S = (1,2,3,4,5,6)^{D_1}$

D_1 D_2 D_3 D_4 (Number)

Prob.



Case (A) $a = b = c = d$ $\quad D_1 = D_2 = D_3 = D_4$

Fav Choices $= 6 \times 1 \times 1 \times 1$

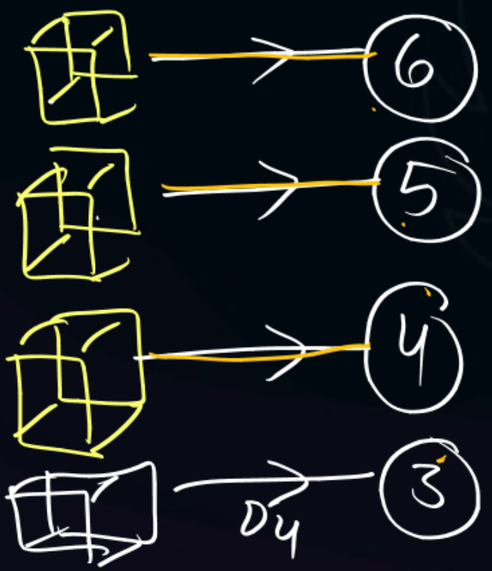
$$P(E) = \frac{6 \times 1 \times 1 \times 1}{6 \times 6 \times 6 \times 6} \quad \text{— (1)}$$

Case 02 $a \neq b \neq c$

$d \leftarrow a, b, c$

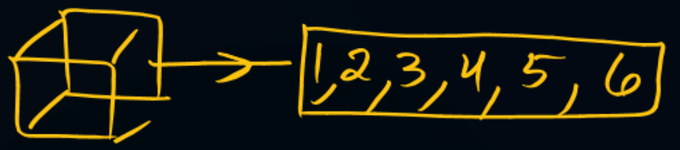
D_4 shares one of D_1, D_2, D_3

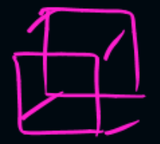
$\equiv$ $D_4 \leftarrow a, b, c$



$\Rightarrow \dfrac{6 \times 5 \times 4 \times 3}{6 \times 6 \times 6 \times 6}$ — (2)

$a = b = c = d$







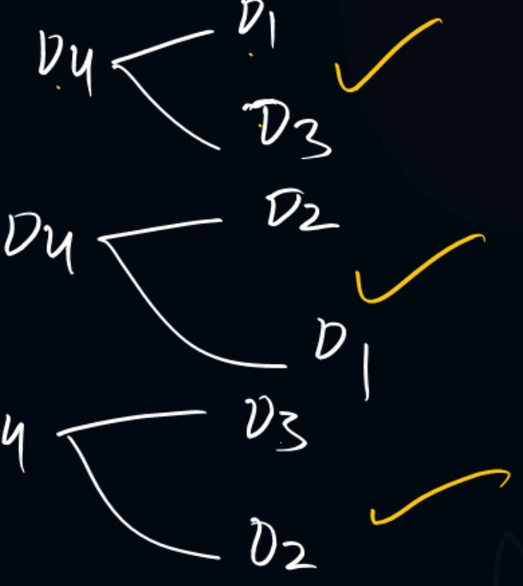


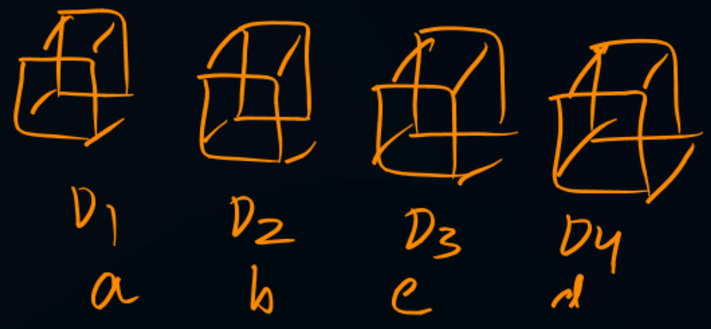
CASE 03

$\frac{6\times1\times5\times2}{6\times6\times6\times6}\times3$ $\Rightarrow$ $a=b\neq c$, $D_1=D_2\neq D_3$

$\Rightarrow$ $b=c\neq a$, $D_2=D_3\neq D_1$

$\Rightarrow$ $c=a\neq b$, $D_3=D_1\neq D_2$





$$\Rightarrow \frac{6\times1\times5\times2}{6\times6\times6\times6}\times3$$

$$P(E)=\frac{6\times1\times1\times1}{6\times6\times6\times6}+\frac{6\times5\times4\times3}{6\times6\times6\times6}+\frac{6\times1\times5\times2}{6\times6\times6\times6}\times3=\frac{91}{216}$$

Classification of Events :

compound events ✓ Min 2 events
✓ more Than 2 events

single event $P(E) = \frac{n(E)}{n(S)}$

↓ more Than 2 events

CASE(A) something is common

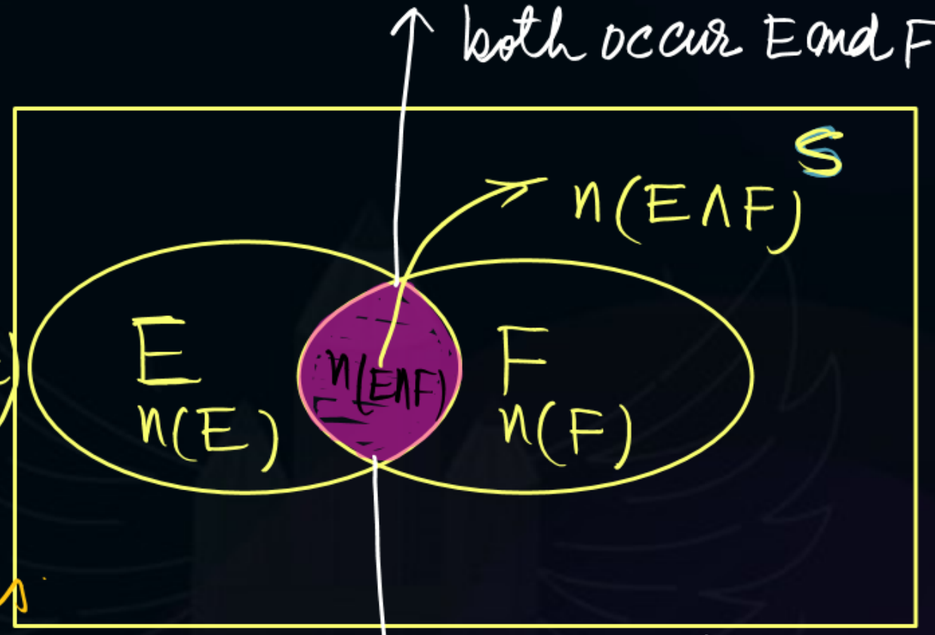
$P(E \wedge F) = P(E \text{ and } F)$

$= P(\text{common portion}) = P(\text{both occur})$

$P[E \wedge F] = \frac{\text{No. of favourable}}{\text{Possible No. of Total outcomes}}$

$P(E \text{ and } F)$

$$P(E \wedge F] = \frac{n(E \wedge F)}{n(S)}$$



What is the Prob.

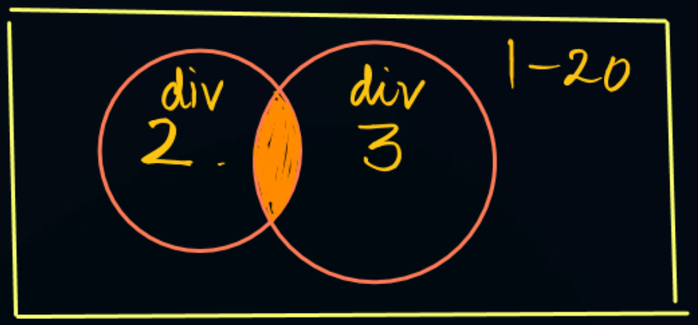
$P(\text{div by 2 and 3}) = P(2 \wedge 3)$

$$P(2 \wedge 3) = \frac{n(2 \wedge 3)}{n(S)} = \frac{3}{20}$$

$$\boxed{P(2 \wedge 3) = \frac{3}{20}}$$

$P(\text{both Happening / 2 and 3 / common})$

$$\boxed{P(2 \wedge 3) = \frac{3}{20}}$$



div by 2 = 2, 4, 6, 8, 10, 12, 14, 16, 18, 20

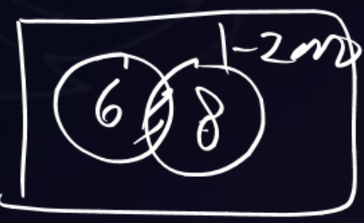
div by 3 = 3, 6, 9, 12, 15, 18,

Common element = 3

$n(E \wedge F) = 3$

$\frac{20}{6} = 3$

$\frac{200}{24} = 8$





✓ CASE 02 (Something is common)

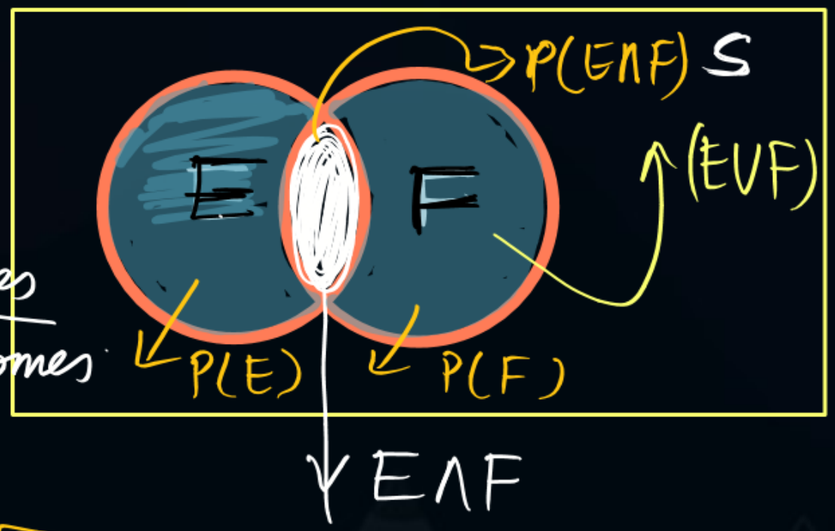
$$P(E \text{ or } F) = P(E \cup F) = P(E \text{ or } F)$$

$$P(E \cup F) = \frac{\text{No. of favourable outcomes}}{\text{Total No. of Possible outcomes}}$$

$$P(E \cup F) = \frac{n(E \cup F)}{n(S)}$$

$$P(E \cup F) = \frac{n(E) + n(F) - n(E \cap F)}{n(S)}$$

$$P(E \cup F) = \frac{n(E)}{n(S)} + \frac{n(F)}{n(S)} - \frac{n(E \cap F)}{n(S)}$$



$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

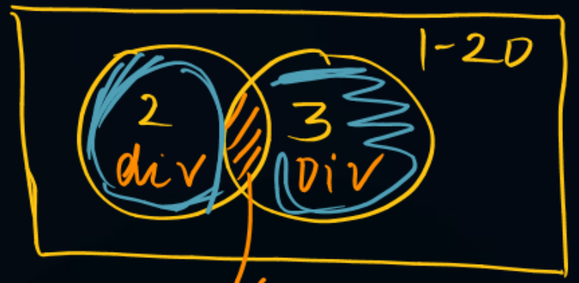
Prob. Addition Theorem

What is the Prob. $P(\text{div by } 2 \text{ or } 3)$

$P(E \cup F) = P(E) + P(F) - P(E \cap F)$

$P(2 \cup 3) = P(2) + P(3) - P(2 \cap 3)$

$= \frac{10}{20} + \frac{6}{20} - \frac{3}{20} = \frac{13}{20}$



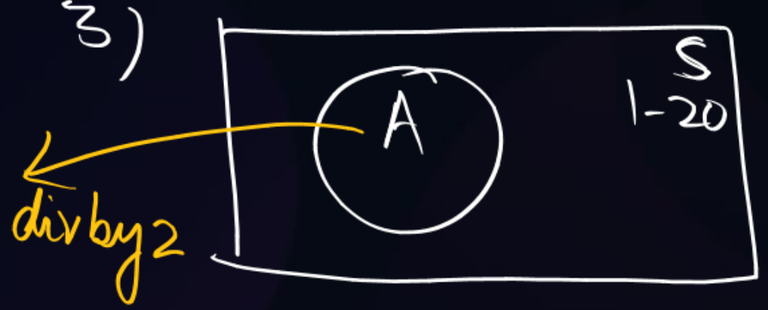
$L(2 \cap 3)$

$n(S) = 20$

$P(2) = \frac{n(2)}{n(S)} = \frac{(2, 4, 6 \cdots 20)}{20}$

$P(2 \cup 3) = \frac{13}{20}$

3)



What is the Prob.

$P(\text{Not divisible by } 2)$

Wrong Rule of Prob:

$P(\text{div by 2}) + P(\text{Not div 2}) = 1$

$= \boxed{P(A) + P(\bar{A}) = 1}$

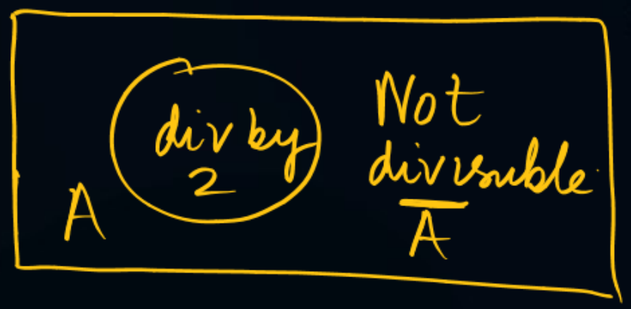
$= P(\text{Not div by 2}) = 1 - P(\text{div by 2})$

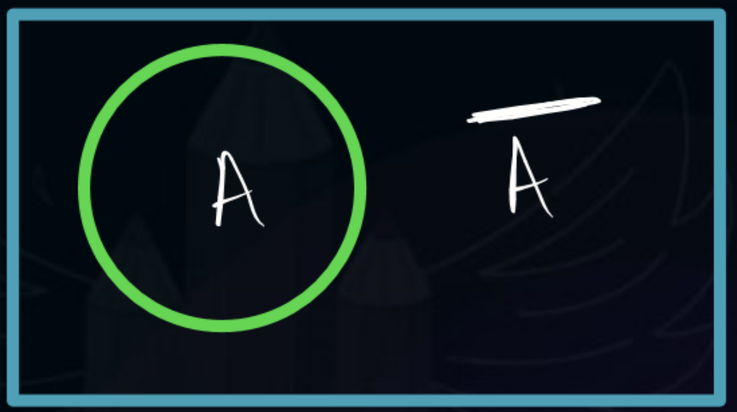
$= 1 - P(A)$

$= 1 - \frac{10}{20}$

$= \frac{10}{20}$ ✓

$\boxed{P(\bar{A}) = \frac{10}{20}}$



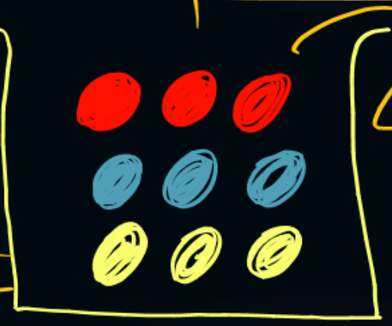


- ✓ $P(E \text{ and } F) = \cap$
- ✓ $P(E \text{ or } F) \quad \cup$
- ✓ $P(\bar{E}) \quad 1 - P(E)$

Dependent events: (without replacement)

What is the Prob P(3 red balls Are drawn at random one at a time)

both occur
red 1 ∧ red 2



I st red ball

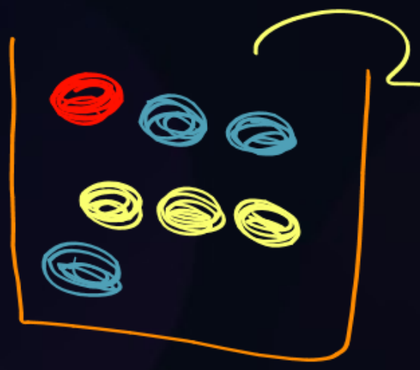
$P(\text{Red } 1) = \frac{3}{9}$

R_2 / R_1 R_1 is already occured
R_2 occuring



II nd red ball

$P\left(\frac{\text{Red } 2}{\text{Red } 1}\right) = \frac{2}{8}$



III rd red ball

$P\left(\frac{\text{Red } 3}{\underbrace{\text{Red } 1 \wedge \text{Red } 2}_{\text{both}}}\right) = \frac{1}{7}$

$$P(\text{red 1}) = \frac{3}{9} \quad P\left(\frac{\text{Red 2}}{\text{Red 1}}\right) = \frac{2}{8} \quad P\left(\frac{\text{Red 3}}{\text{Red 1} \wedge \text{red 2}}\right) = \frac{1}{7}$$

$P(\text{Red 1} \wedge \text{red 2} \wedge \text{red 3})$ Three working Together

$$P(R_1 \wedge R_2 \wedge R_3) = P(R_1)\, P\left(\frac{R_2}{R_1}\right)\, P\left(\frac{R_3}{R_1 \wedge R_2}\right)$$

If $R_1 \longrightarrow A$
$R_2 \longrightarrow B$
$R_3 \longrightarrow C$

$$P(A \wedge B \wedge C) = P(A)\, P\left(\frac{B}{A}\right)\, P\left(\frac{C}{A \wedge B}\right)$$

If Two Events Are Dependent

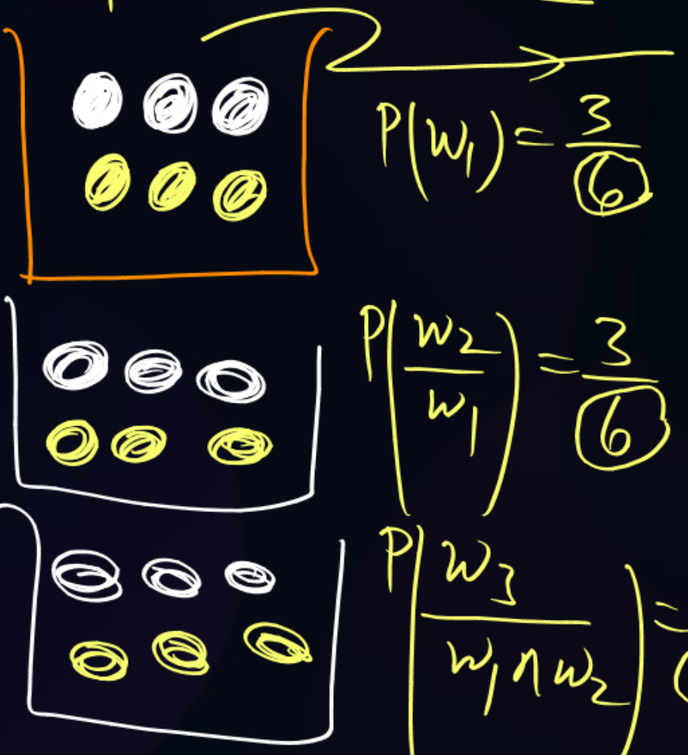
$$\boxed{P(A \wedge B) = P(A)\, P\left(\frac{B}{A}\right)}$$

for n events

Dependence of Events

$$P(A \cap B \cap C \cap D \cap E \cap G \cap \dots) = P(A)\,P\left(\frac{B}{A}\right)\,P\left(\frac{C}{A \cap B}\right)\,P\left(\frac{D}{A \cap B \cap C}\right)\dots$$

Independent events : with replacement : P(3 White ball are Drawn)



$$P(W_1) = \frac{3}{6}$$

$$P\left(\frac{W_2}{W_1}\right) = \frac{3}{6}$$

$$P\left(\frac{W_3}{W_1 \cap W_2}\right) = \frac{3}{6}$$

$$P(W_1 \cap W_2 \cap W_3) = P(W_1)\,P\left(\frac{W_2}{W_1}\right)\,P\left(\frac{W_3}{W_1 \cap W_2}\right)$$

$$P\left(\frac{W_2}{W_1}\right) = P(W_2)$$

$$P\left(\frac{W_3}{W_1 \cap W_2}\right) = P(W_3)$$

$$P(W_1 \cap W_2 \cap W_3) = P(W_1)\,P(W_2)\,P(W_3)$$

If $W_1 = A \quad W_2 = B \quad W_3 = C$

$$\boxed{P(A \cap B \cap C) = P(A)\,P(B)\,P(C)} \Longrightarrow \text{"Independence of events"}$$

If Two events Are Independent

$$\boxed{P(A \cap B) = P(A)\,P(B)}$$

If 'n' Independent events

$$\boxed{P(A \cap B \cap C \cap D \cap E \cap F \cap G \text{--}) = P(A)\,P(B)\,P(C)\,P(D) \text{--}}$$

dependent

$$P(W_1 \cap W_3) = P(W_1)\,P\left(\frac{W_3}{W_1}\right)$$

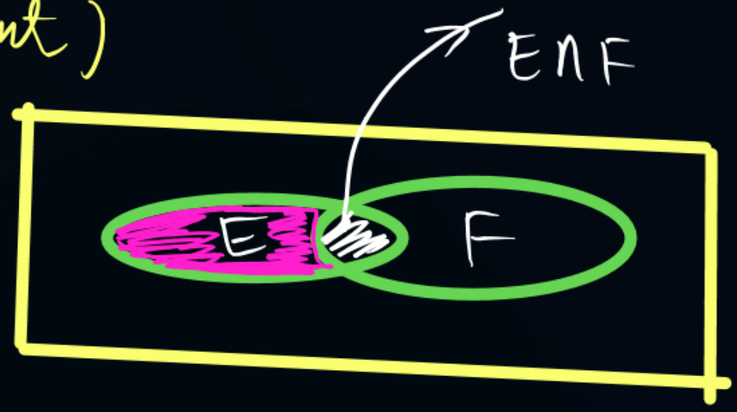
$$P(W_1 \cap W_3) = P(W_1)\,P(W_3)$$

Independent

CASE 05 If Two Events E and F (both are Independent)

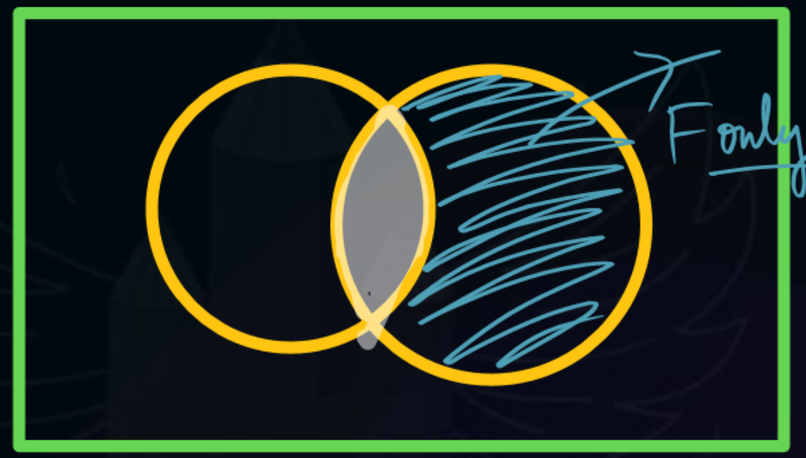
$P(\text{only } E) = P(E \wedge \bar{F}) = \text{only } E$

E and F Are Independent events
$\bar{E}$ and $\bar{F}$ Are Independent events]



$$P(\text{only } E) = P(E \wedge \bar{F}) = P(E)\,P(\bar{F})$$

$$\boxed{P(\text{only } E) = P(E \wedge \bar{F}) = P(E)\,[1 - P(F)]}$$



$P(\text{only } F) = P(\bar{E} \wedge F) = P(\bar{E})\,P(F)$

$$P(\text{only } F) = [1 - P(E)]\,P(F)$$

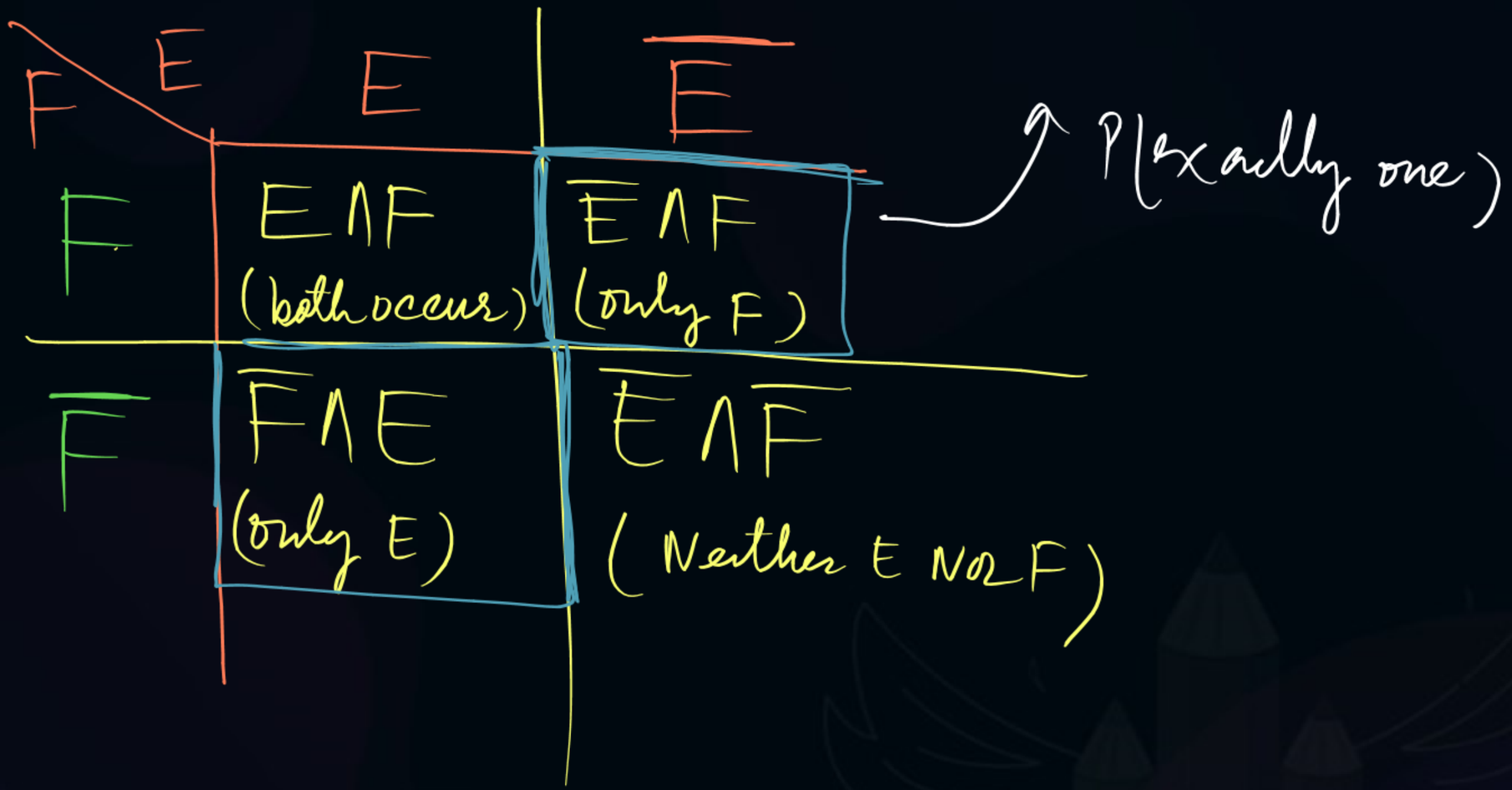
If E and F are Independent events

$$P(\text{exactly one}) = P(\text{only } E) + P(\text{only } F)$$

$$= P(E \cap \bar{F}) + P(F \cap \bar{E})$$

$$\boxed{P(\text{exactly one}) = P(E) + P(F) - 2P(E)P(F)}$$

$$P[(E \cap \bar{F}) \cup (F \cap \bar{E})]$$


E
$\bar{E}$
F
$E \cap F$
(both occur)
$\bar{E} \cap F$
(only F)
$\bar{F}$
$F \cap \bar{E}$
(only E)
$\bar{E} \cap \bar{F}$
(Neither E NOR F)
P(exactly one)

THANK - YOU